Improving Named Entity Recognition in Song Lyrics Through Domain-Adaptive Pretraining and Genre-Specific Fine-Tuning

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Abstract

Creating a Named Entity Recognition (NER) model that performs well on domain-specific datasets can be very challenging, especially since most NER models are trained on formal or general-purpose data without being tailored to a particular domain. In this project, we focus on a dataset of song lyrics. Song lyrics differ significantly in language style, often including creative spelling, slang, abbreviations, and pop culture references—posing a considerable challenge for entity recognition tasks.

In this paper, we explore how to improve the performance of an NER model originally trained on general-purpose data by applying domain-adaptive pretraining, fine-tuning, and continuous learning using lyrics from three music genres: pop, country, and rap/hip-hop. We compare several models and evaluate their contribution to performance improvements. Our results show that domain adaptation significantly outperforms the baseline general-purpose model, achieving an improvement of 7.19 percentage points in span-F1 score. These findings highlight the importance of domain adaptation for robust entity recognition in creative and informal datasets. [1]

1 Introduction

In Named Entity Recognition (NER) models, the accurate identification of named entities is crucial. For optimal performance, these models must effectively capture the unique characteristics of the data they are trained on. When data, such as song lyrics, contain distinct features, such as informal language, slang, references to well-known figures and brands, and poetic or abstract expressions, traditional NER models trained on general-purpose datasets may struggle to capture these nuances. As a result, their performance may lack robustness and accuracy.

Techniques such as domain-adaptive pretraining and fine-tuning have been proven to improve the performance of NER models, but how effective are these techniques when applied to song lyrics? Specifically, to what extent does domain-adaptive pretraining and fine-tuning on song lyrics enhance NER model performance compared to general-purpose model, and do certain music genres contribute more effectively to this adaptation?

2 Related Work

Prior work has demonstrated the effectiveness of domain-adaptive pretraining (DAPT) and task-adaptive pretraining (TAPT) in improving performance on downstream NLP tasks. Gururangan et al. (2020) show that continued pretraining in unlabeled domain-specific data significantly boosts performance, even when using large-scale models such as RoBERTa. Sachidananda et al. (2021) further highlight the importance of adapting tokenization and pretraining to domain-specific lexical properties, especially in multilingual contexts. Although these studies target formal or technical domains, such as biomedical or financial texts, our work focuses on a significantly underexplored domain: song lyrics. This domain poses unique challenges for NER due to its creative use of language, informal tone, and frequent use of slang and cultural references. Although prior research by Song and Beck (2023) has explored the modeling of emotions in lyrics, they do not involve structured NER tasks or manually annotated data. Similarly, Hong et al. (2023) explore domain relevance via masked domain-specific keywords, yet still operate on

[1] https://github.com/aniela2906/NLP-spring-2025.git

large, structured corpora and do not address annotation or evaluation in informal domains. In contrast, we manually annotate a novel dataset of song lyrics with standard NER tags, and systematically evaluate both general-purpose and lyrics-specific models. To our knowledge, this is the first work that directly compares general and domain-specific NER models on a manually annotated song lyrics dataset.

3 Data and Annotation

3.1 Data Collection

To construct our datasets, we first identified popular artists in three music genres—pop, country, and rap/hip-hop—using rankings from Kworb, Vault, and Browne et al. (2017). We then selected 50 artists per genre, and for each artist, we chose 20 songs based on popularity from Kworb and lyric availability on Genius (b). We scraped the lyrics for all selected songs using Genius open API (Genius, a), resulting in 1,000 songs per genre. We preprocessed the lyrics by removing duplicate or repetitive lines commonly found in songs. Finally, the lyrics were tokenized and converted into the IOB2 format.

3.2 Manual Annotation

Since many sentences in our datasets did not contain any named entities, we first used our DAPT model trained only on the English Web Treebank (EWT) dataset (Mayhew, 2022b) (this model is described in section 4.1) to automatically label the data and identify sentences likely to include named entities. Based on this initial labeling, we sampled and manually annotated 2,000 sentences from each music genre dataset (pop, country, and rap/hip-hop), following the annotation guidelines provided by Mayhew (2022a). The annotation scheme was consistent with that of the EWT dataset, using the entity tags: B-PER, I-PER, B-ORG, I-ORG, B-LOC, I-LOC, and O. After annotation, the data were shuffled and randomly split into training and test sets. Each training file contains 1,000 sentences per genre, while the test set consists of 3,000 sentences (1,000 from each genre). In addition, we created a combined training file that included all three genres. While sentences from the same song may appear in both the training and test sets, we ensured that no repeated or identical lines were present through preprocessing. Given that our model operates at the sentence level without broader song-level context, this does not result in data leakage or inflated performance metrics.

3.3 Exploratory Data Analysis

To better understand the characteristics of our data, we conducted an exploratory analysis of both general-purpose and genre-specific datasets. As shown in Table 1, the EWT dataset used for baseline training contains over 229,000 tokens, but a relatively low proportion of named entity tokens (5.16%). In contrast, our manually annotated genre-specific datasets—each comprising 1,000 sentences from pop, country, and rap/hip-hop lyrics—exhibit significantly higher entity densities: country leads with 16.9%, followed by rap/hip-hop (15.3%) and pop (12.4%). This is a result of intentionally sampling sentences that contain named entities for the training sets. The pseudo-labeled datasets contain far more sentences and tokens overall, but have lower entity densities, with rap/hip-hop showing noticeably higher density than pop or country (pseudo-labeling is described in section 4.2). Finally, our manually annotated lyrics test set, consisting of 3,000 sentences, has a high entity density of 14.83%, making it a robust benchmark for evaluating model performance. Table 2 further shows the distribution of unique named entities across datasets, with rap/hip-hop containing significantly more unique entities than the other genres.

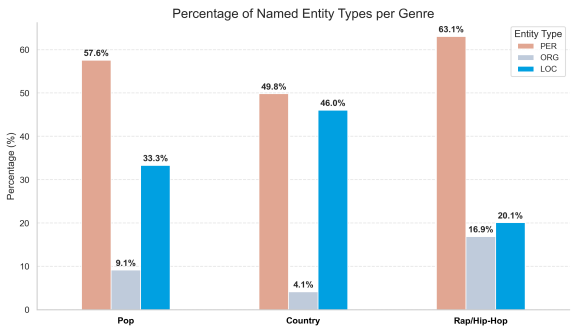

Figure 1: Percentage of each named entity in different genres.

Figure 1 breaks down the distribution of named entity types across genres more. For additional

Dataset	Sentences	Tokens	Named Entity Tokens	Entity-Labeled Token Density (%)	Avg Tokens/Sent.
EWT data	14,544	229,728	11,856	5.16	15.80
pop (manually labeled)	1,000	9,295	1,153	12.4	9.29
country (manually labeled)	1,000	10,167	1,718	16.9	10.17
rap/hip-hop (manually labeled)	1,000	11,439	1,750	15.3	11.44
pop (pseudo-labeled)	33,129	318,566	484	0.15	9.62
country (pseudo-labeled)	18,418	175,577	728	0.41	9.53
rap/hip-hop (pseudo-labeled)	51,141	577,204	11,630	2.01	11.29
lyrics test	3,000	31,094	4,610	14.83	10.36

Table 1: **S**ummary of datasets' characteristics.

context, a bar chart illustrating the absolute counts of each named entity type can be found in the Appendix (see Figure 4). In pop lyrics, person (PER) entities dominate at 57.6%, followed by locations (LOC, 33.3%) and a small part of organizations (ORG, 9.1%). Country lyrics show a more balanced entity profile, with 49.8% PER and a notably high 46.0% LOC entities, reflecting the genre's frequent geographic references. In contrast, rap/hip-hop lyrics are heavily skewed toward PER entities (63.1%), with fewer LOC (20.1%) but a relatively higher proportion of ORG entities (16.9%), possibly reflecting references to brands, labels, or groups. These findings highlight the distinct thematic emphases of each genre and reinforce the importance of genre-specific fine-tuning for effective named entity recognition in lyrical text.

Dataset	PER	ORG	LOC	Unique Named Entity Tokens
EWT data	3,217	1,660	3,111	3390
pop (manually labeled)	510	72	244	499
country (manually labeled)	585	38	528	532
rap/hip-hop (manually labeled)	772	188	235	809
pop (pseudo-labeled)	248	73	59	202
country (pseudo-labeled)	393	64	112	200
rap/hip-hop (pseudo-labeled)	4,511	1,872	1,366	4012
lyrics test	1,807	295	1,066	1704

Table 2: Number of specific named entity tokens in each dataset.

4 Methodology

4.1 Baseline Model

Our baseline model is based on deepset/roberta-base-squad2, a RoBERTa transformer pre-trained for question answering (Liu et al. (2019), Wolf et al. (2020)). We selected this model because its span-sensitive training aligns well with NER objectives, and our preliminary experiments showed it outperformed the standard RoBERTa on the EWT dataset. We adapted the model for token classification using standard NER supervision, including sentence-level tokenization, BIO tagging, and alignment of labels to word-piece tokens. Results for EWT test data can be found in the Appendix (Figure 3).

To strengthen our in-domain baseline, we applied DAPT, which involved masked language modeling (MLM) on a large corpus of song lyrics prior to fine-tuning the model for NER. All subsequent models (pop, country, rap/hip-hop, and the combined three-genre model) were initialized from this DAPT model, ensuring a consistent starting point and enabling fair comparisons across training strategies.

4.2 Fine-Tuning

We fine-tuned our DAPT model, trained on the EWT dataset, separately on three genre-specific datasets (pop, country, rap/hip-hop) and one combined dataset, resulting in four distinct models. Next, we used the resulting fine-tuned models to pseudo-label the remaining, unlabeled song lyrics - each model labeling data from its corresponding genre. We then conducted an additional round of fine-tuning using both the manually labeled and pseudo-labeled data for each genre and the combined set. At each stage of this pipeline, we evaluated performance on the lyrics test set to monitor potential improvements in the performance.

5 Results

To evaluate the performance of different NER models on a dataset with significant class imbalance (the overwhelming majority of "O" tags and unequal distribution of named entity types), relying on accuracy alone would be misleading. Therefore, we use the widely accepted span-F1 score, which balances precision and recall via the harmonic mean. This metric considers a prediction correct only if both the entity type and the entire span exactly match the gold annotation.

Table 3 presents the evaluation results (bar

chart version can be found in the Appendix, see Figure 5). All models were evaluated on the same combined lyrics test set, regardless of which genre the model was trained on. The baseline model, trained only on the EWT dataset, achieved a span-F1 score of 80.28%. The DAPT model, pretrained on the full corpus of lyrics, showed a noticeable improvement with 81.37%. All models fine-tuned on manually annotated song lyrics showed significant gains, with the rap/hip-hop model achieving the highest score among the single-genre models (85.19%), followed by pop (85.08%) and country (84.65%). The model trained on all manually labeled lyrics across genres achieved the highest overall score of 87.46%.

Applying continuous learning using pseudo-labeled data led to modest improvements across all models, with increases of less than one percentage point (a file containing detailed results for each model is available on our GitHub).

To assess whether these small gains from continuous learning were statistically significant, we conducted a paired bootstrap significance test with 1,000 resamples. Only the rap/hip-hop model showed a significant improvement ($p<0.001$), while results for pop ($p=0.109$), country ($p=0.061$), and the combined three-genre model ($p=0.614$) were not significant.

Train Dataset	Span-F1 Score (%)	Unique NE Coverage (%)
EWT (basic)	80.28	13.67
EWT (DAPT)	81.37	13.67
EWT (DAPT) + pop	85.08	23.06
EWT (DAPT) + pop (cont.)	85.51	24.06
EWT (DAPT) + country	84.65	25.88
EWT (DAPT) + country (cont.)	85.09	26.88
EWT (DAPT) + rap/hip-hop	85.19	23.88
EWT (DAPT) + rap/hip-hop (cont.)	85.85	38.32
EWT (DAPT) + all genres	87.46	41.14
EWT (DAPT) + all genres (cont.)	87.47	53.11

Table 3: Span-F1 score results for all models and their coverage.

6 Analysis

As domain-specific data was progressively added, the unique named entity coverage in the test data significantly increased (see Table 3). The coverage rose from 13.67% in models trained exclusively on the EWT dataset, to 23.06% in pop model, 25.88% country model, and 23.88% in rap/hip-hop model. Further incorporating pseudo-labeled data led to only slight increases in coverage for pop and country, as these datasets contained fewer named entity tokens. However, the rap/hip-hop model saw a substantial increase, with coverage reaching 38.32%. This improved coverage is generally reflected in the models' span-F1 scores (see Table 3)

As expected, the set combining all genres yielded the highest coverage at 41.14%, with its continuous learning version achieving an impressive 53.11%. These results highlight the importance of domain adaptation. Clearly, song lyrics contain many named entities that are uncommon in universal datasets like EWT, which explains the marked improvement in coverage as more domain-specific data was added. However, it is essential to note that while allowing the model to learn specific named entities is beneficial, it does not guarantee that the model will generalize well to unseen data, especially when dealing with differently structured sentences.

To better understand differences in models' performance, we analyzed the distribution of error types across models (Figure 2). We focused on three error types: missed entities (named entities not detected), spurious entities (non-entities incorrectly labeled as entities), and wrong labels (entities detected but misclassified). The most notable improvement occurs with missed entities, which drop significantly after introducing domain-specific data - from 791 in the EWT-only model to 210 in the country model. This may be due to the model being exposed to song-specific named entities during training, increasing its tendency to recognize similar ones in the test set.

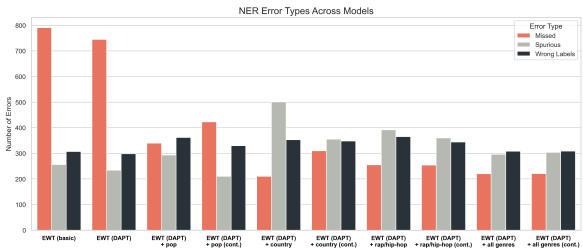

Figure 2: NER error types across models.

For both the pop and country models, adding pseudo-labeled data shows a tradeoff: while the number of spurious errors decreases, the number of missed entities increases. Interestingly, this trend is not observed in the rap/hip-hop model or in the all genres combined model, where the errors behave differently. The wrong labels error type seems to vary the least across models. In most cases, models either fail to detect a named entity or, conversely, are too generous with labeling.

These observations are quite insightful, as they suggest that while continuous learning did not significantly improve performance, except in rap/hip-hop model, it did alter the distribution of errors. A more detailed breakdown of the relative proportions of each error type across models is provided in the Appendix (see Figure 6)

7 Discussion

Both DAPT and fine-tuning on domain-specific data significantly improved model performance, yielding a 7.18 percentage point gain in span-F1 score. In contrast, adding pseudo-labeled data led to only marginal improvements, and these gains were statistically significant only for the rap/hip-hop model. This suggests that the most impactful factors for improving performance are learning the structure of domain-specific language and access to high-quality annotated data.

These findings directly address our research question, demonstrating that domain-adaptive pretraining and genre-specific fine-tuning meaningfully enhance NER performance compared to a general-purpose model. While all three music genres exhibited similar performance trends, rap/hip-hop stood out in its response to pseudo-labeled data, suggesting that both the quantity and diversity of named entities in that genre play a crucial role.

It is important to note that the three genres differ substantially in terms of sentence structure and the density and diversity of named entities. While we specifically sampled sentences containing named entities for annotation, raw unfiltered lyrics reveal that rap/hip-hop inherently features more named entities, and they tend to be more diverse than in other genres. This makes rap/hip-hop a particularly rich source for NER tasks. Moreover, rap/hip-hop songs tend to be longer and contain more sentences per track compared to pop or country, making it easier to collect a large and diverse dataset from fewer sources.

Another important factor is the manual annotation process itself. Labeling was performed entirely by us, which is both time-consuming and susceptible to human error. This inherently limits the size of the datasets we could use for training and testing. Additionally, ambiguity in the lyrics presented challenges during annotation. To address this, we established shared annotation guidelines beforehand and collaboratively resolved ambiguous cases during the labeling process to ensure consistency. This was especially important for rap/hip-hop, where nicknames, slang, and non-standard language forms are prevalent, and domain knowledge is required.

8 Conclusion

This study highlights the importance of domain-specific adaptation in natural language processing tasks, particularly for NER in the context of song lyrics. Our work shows that DAPT followed by genre-specific fine-tuning significantly improves model performance, making a span-F1 score increase of up to 7.19 percentage points compared to the baseline model trained only on general-purpose data. The model trained on all three music genres combined performed best, achieving the highest F1 score and the broadest named entity coverage. All genres contributed comparably to performance gains, with rap/hip-hop showing slightly better results, likely due to its dense and diverse use of named entities. These findings prove that focusing on domain-specific NLP is crucial when dealing with informal, creative datasets like song lyrics, and that training models on the right kind of text and labeling data accurately can lead to strong and reliable NER systems that understand the context better.

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Appendix

Group Contributions: All group members contributed equally to the project. The manual annotation task was divided among members, with each person labeling 2,000 sentences from a specific music genre. The group collaborated on all other aspects of the project, distributing tasks evenly and reviewing each other's work to ensure accuracy and quality.

Evaluation results on the **English Web Treebank (EWT) test set**:

recall: 0.8591160220994475
precision: 0.8536139066788655
span-f1: 0.8563561266636073

Figure 3: Baseline model results on EWT test set.

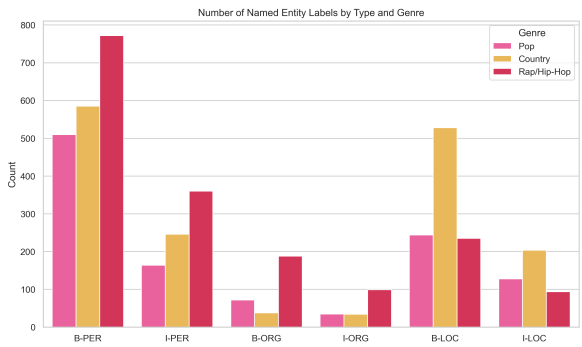

Figure 4: Number of named entity labels by type and genre.

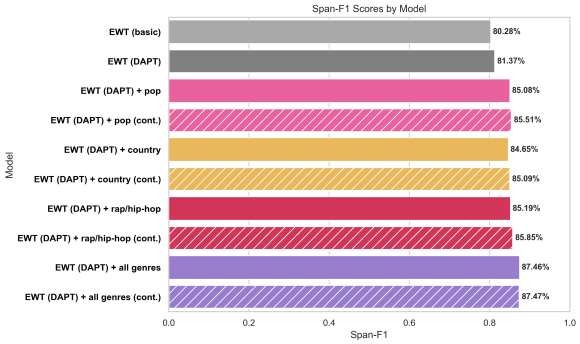

Figure 5: Span-F1 score for each model.

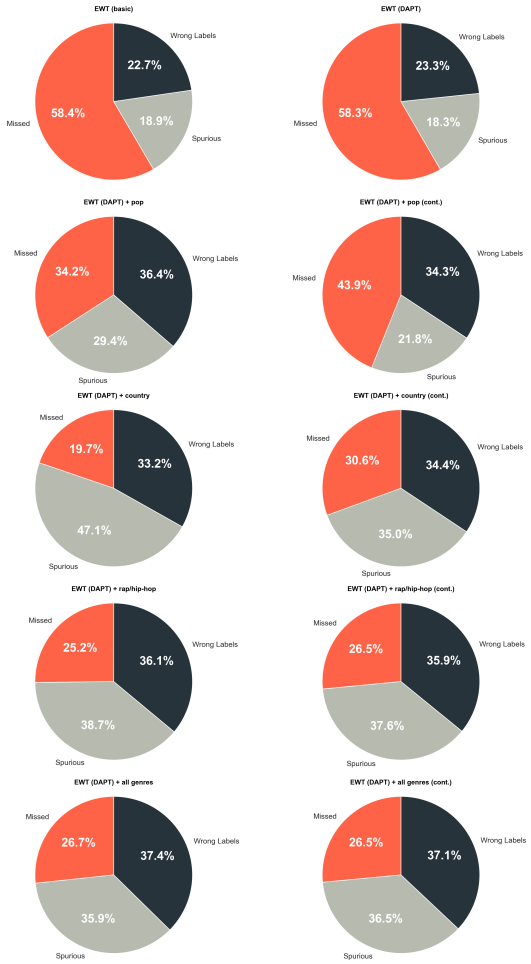

Figure 6: Percentage of each error type in all models.